Azmari Sultana

TOPIC NAME : Random Forest DAY :

TIME : DATE : / /

3 bootstrapped dataset with 6 instances each:

Bootstrapped dataset 1:

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	overcast	Mild	High	Strong	Yes
Day13	overcast	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	No
Day2	Sunny	Hot	High	Strong	No

Bootstrapped dataset 2:

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
Day 1	Sunny	Hot	High	Weak	No
Day 2	Sunny	Hot	High	Strong	No
Day 3	overcast	Hot	High	Weak	Yes
Day 4	Rain	Mild	High	Weak	yes
Day 5	Rain	Cool	Normal	Weak	yes
Day 2	Sunny	Hot	High	Strong	No

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Bootstrapped dataset 3 :

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
Day 6	Rain	Cool	Normal	Strong	No
Day 7	Overcast	Cool	Normal	Strong	yes
Day 8.	Sunny	Mild	High	Weak	No
Day 9	Sunny	Cool	Normal	weak	yes
Day 10	Rain	Mild	Normal	weak	yes
Day 3	Overcast	Hot	Normal	weak	yes

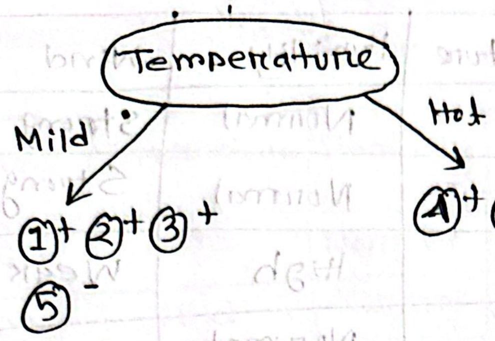
Bagging → create several modes on different random samples (bootstrapped) dataset.

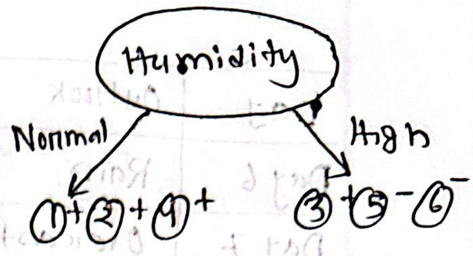
Feature selection → consider only a subset of features instead of all.

For bootstrapped dataset 1 :

Day	Temperature	Humidity	Play Tennis
Day 10	Mild	Normal	yes
Day 11	Mild	Normal	yes
Day 12	Mild	High	yes
Day 13	Hot	Normal	yes
Day 14	Mild	High	no
Day 2	Hot	High	no

Calculating Gini Impurity :





$$\text{Gini(Hum)} = \frac{3}{6}\left(1-\left(\frac{3}{3}\right)^2-\left(\frac{0}{3}\right)^2\right) + \frac{3}{6}\left(1-\left(\frac{1}{3}\right)^2-\left(\frac{2}{3}\right)^2\right)$$

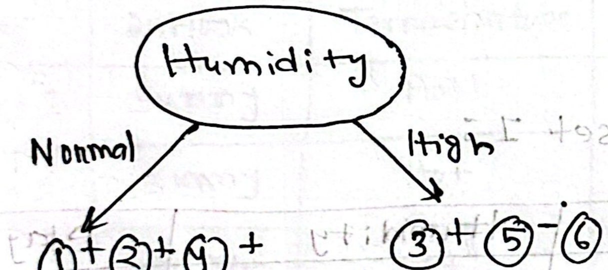
$$= 0.232$$

$$\text{Gini(temp)} = \frac{4}{6}\left(1-\left(\frac{3}{4}\right)^2-\left(\frac{1}{4}\right)^2\right) + \frac{2}{6}\left(1-\left(\frac{1}{2}\right)^2-\left(\frac{1}{2}\right)^2\right)$$

$$= 0.417$$

Gini (Temp) > Gini (Humidity)

So root node = Humidity



Decision = yes (pure)

(impure)

we need to again decide feature for this node as split.

For impure node, we have to do the same process again. so we'll seperate the datapoints of the impure node from Bootstrapped dataset 1.

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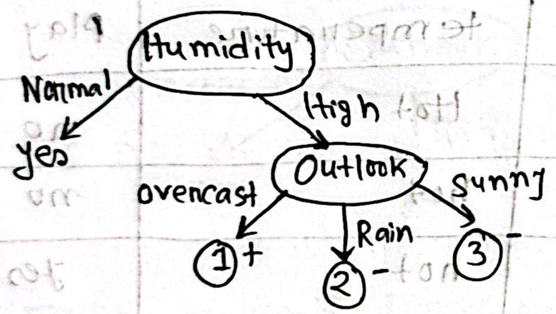
Data set for the next level Node :

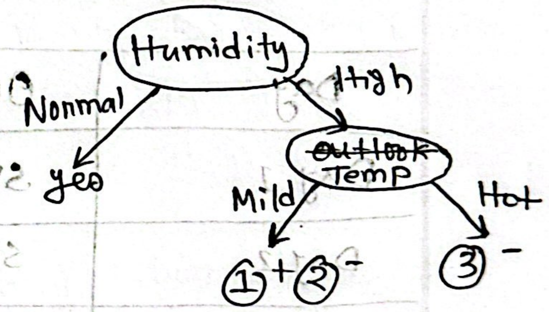
Day	Outlook	Temperature	Humidity	Wind	Play tennis
Day12	Overcast	Mild	high	Strong	yes
Day14	Rain	Mild	high	Strong	no
Day 2	sunny	hot	high	strong	no

Again we'll select two features :

Day	outlook	Temperature	Play tennis
Day 12	Overcast	Mild	yes
Day 14	Rain	Mild	no
Day2	Sunny	Hot	no

Gini Impurity :





$$\text{Gini}(\text{outlook}) = \frac{1}{3}\left(1-\left(\frac{1}{1}\right)^2\right) + \frac{1}{3}\left(1-\left(\frac{1}{1}\right)^2\right) + \frac{1}{3}\left(1-\left(\frac{1}{1}\right)^2\right)$$

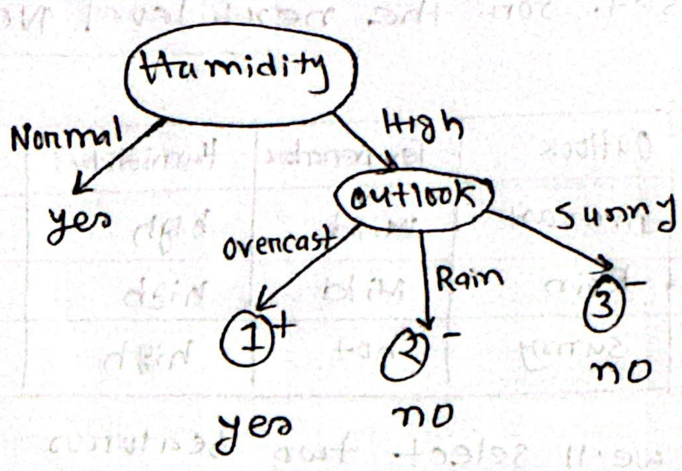
$$= 0$$

$$\text{Gini}(\cancel{\text{Humidity}}\ \text{Temp}) = \frac{3}{3}\left(1-\left(\frac{1}{2}\right)^2-\left(\frac{1}{2}\right)^2\right) + \frac{1}{3}\left(1-\left(\frac{1}{1}\right)^2\right)$$

$$= 0.33$$

$$\text{Gini}(\text{outlook}) < \text{Gini}(\cancel{\text{Humidity}}\ \text{Temp})$$

Now we'll choose outlook.



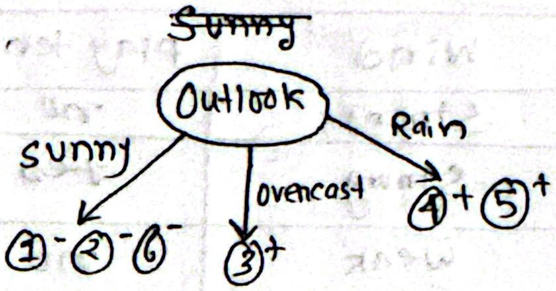
We'll stop growing the tree as all leaves are pure now. So, this is the decesion tree for boot strapped dataset 1.

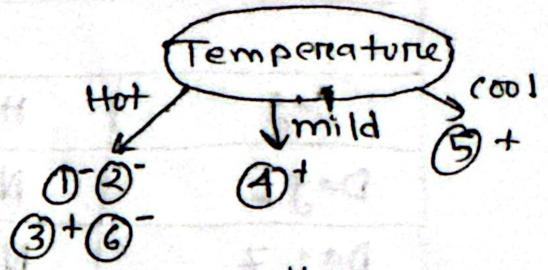
Bootstrapped dataset 2

Day	Outlook	temperature	Play tennis
Day 1	sunny	Hot	no
Day 2	sunny	hot	no
Day 3	overcast	hot	yes
Day 4	Rain	mild	yes
Day 5	Rain	cool	yes
Day 2	Sunny	hot	no

TOPIC NAME : ____________ DAY : ____________ TIME : DATE : / /

Calculating Gini Impurity





$$\text{Gini (temp)} = \frac{4}{6}\left(1-\left(\frac{1}{4}\right)^2-\left(\frac{3}{4}\right)^2\right) + \frac{1}{6}\left(1-\left(\frac{1}{1}\right)^2\right)+\frac{1}{6}\left(1-\left(\frac{1}{1}\right)^2\right)$$

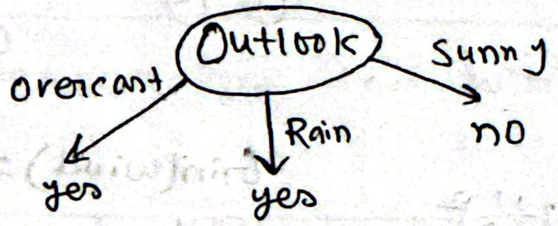
$$= 0.25$$

$$\text{Gini (outlook)} = \frac{3}{6}\left(1-\left(\frac{3}{3}\right)^2\right) + \frac{1}{6}\left(1-\left(\frac{1}{1}\right)^2\right)+\frac{2}{6}\left(1-\left(\frac{2}{2}\right)^2\right)$$

$$= 0$$

Gini(outlook) < Gini (temp)

so root node = outlook



This is the decesion tree for bootstrapped dataset 2

TOPIC NAME : DAY : TIME : DATE : / /

Boot strapped dataset 3

Day	Humidity	Wind	play tennis
Day 6	Normal	Strong	no
Day 7	Normal	Strong	yes
Day 8	High	weak	no
Day 9	Normal	weak	yes
Day 10	Normal	weak	yes
Day 3	Normal	weak	yes

calculating Gini Impurity

Humidity
Normal: 1 − 2 + 4 + 5 + 6 +
High: 3 −

Wind
Strong: 1 − 2 +
weak: 3 − 4 + 5 + 6 +

$$\text{Gini(humidity)} = \frac{5}{6}\left(1-\left(\frac{1}{5}\right)^2-\left(\frac{4}{5}\right)^2\right) + \frac{1}{6}\left(1-\left(\frac{1}{1}\right)^2\right)$$

$$= \frac{4}{15} = 0.267$$

$$\text{Gini(wind)} = \frac{2}{6}\left(1-\left(\frac{1}{2}\right)^2-\left(\frac{1}{2}\right)^2\right) + \frac{4}{6}\left(1-\left(\frac{1}{4}\right)^2-\left(\frac{3}{4}\right)^2\right)$$

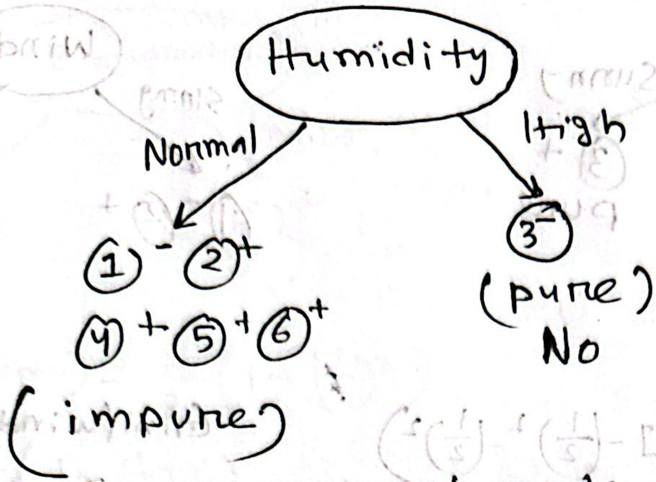
$$= \frac{1}{6} + \frac{1}{4}$$

$$= 0.4167$$

Gini (humidity) < Gini (wind)

Root node = humidity

TOPIC NAME: DAY: TIME: DATE: / /



Data set for next level node

Day	Humidity	Outlook	temperature	wind	Play tennis
Day 6	Normal	Rain	cool	strong	no
Day 7	Normal	overcast	cool	strong	yes
Day 9	Normal	sunny	cool	weak	yes
Day 10	Normal	rain	mild	weak	yes
Day 3	Normal	overcast	hot	weak	yes

Now temperature and wind were selected

Day	~~Temperature~~ Outlook	Wind	Play tennis
Day 6	~~cool~~ rain	Strong	no
Day 7	~~cool~~ overcast	Strong	yes
Day 9	~~cool~~ sunny	weak	yes
Day 10	~~mild~~ rain	weak	yes
Day 3	~~hot~~ overcast	weak	yes

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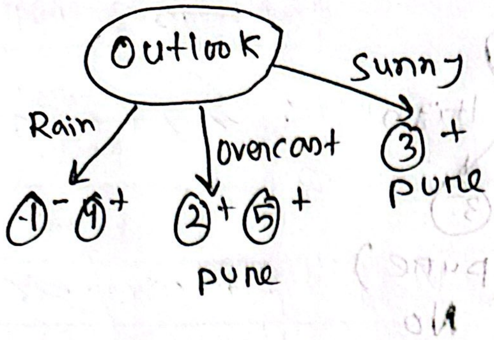
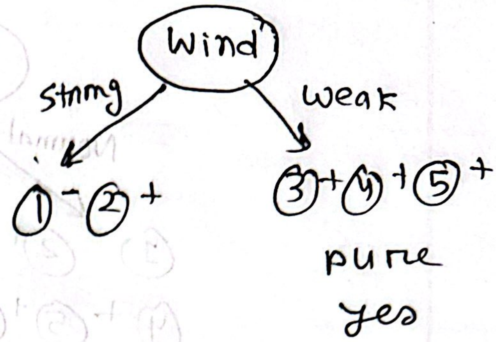




$$Gini(outlook) = \frac{2}{5}\left(1-\left(\frac{1}{2}\right)^2-\left(\frac{1}{2}\right)^2\right) + \frac{2}{5}\left(1-\left(\frac{2}{2}\right)^2\right) + \frac{1}{5}\left(1-\left(\frac{1}{1}\right)^2\right)$$

$$= 0.2$$

$$Gini(wind) = \frac{2}{5}\left(1-\left(\frac{1}{2}\right)^2-\left(\frac{1}{2}\right)^2\right) + \frac{3}{5}\left(1-\left(\frac{3}{3}\right)^2\right)$$

$$= 0.2$$

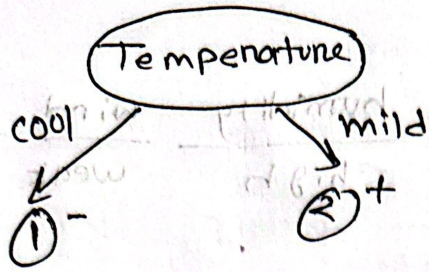
I choose outlook.

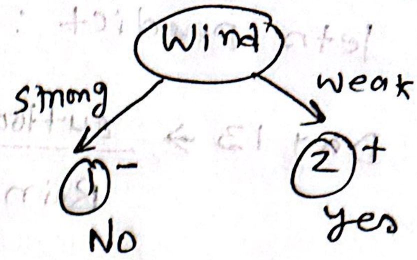
Day	Outlook	temperature	humidity	wind	Play tennis
Day 6	rain	cool	normal	strong	no
Day 10	rain	Mild	normal	weak	yes

Temperature and wind are selected

Day	Temperature	Wind	Play tennis
Day 6	cool	Strong	no
Day 10	mild	weak	yes

TOPIC NAME: DAY: TIME: DATE: / /



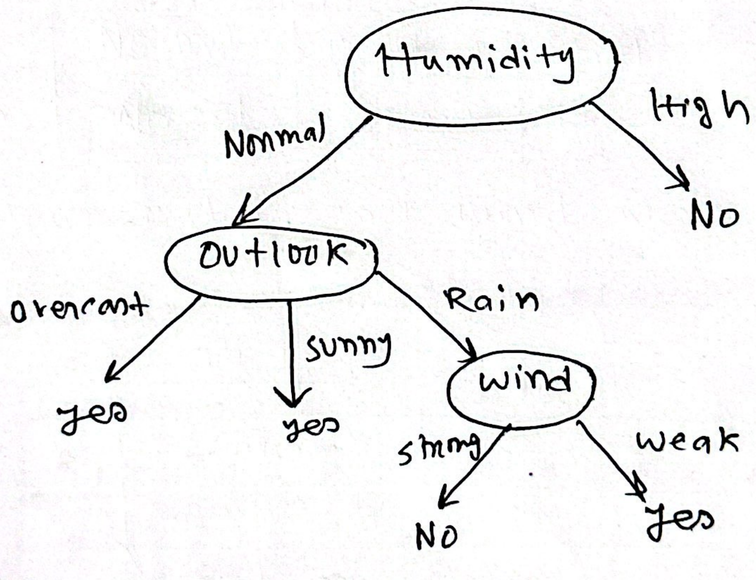


$$Gini(wind) = \frac{1}{2}\left(1-\left(\frac{1}{1}\right)^2\right) + \frac{1}{2}\left(1-\left(\frac{1}{1}\right)^2\right) = 0$$

$$Gini(temp) = \frac{1}{2}\left(1-\left(\frac{1}{1}\right)^2\right) + \frac{1}{2}\left(1-\left(\frac{1}{1}\right)^2\right) = 0$$

$$Gini(temp) = Gini(wind)$$

I choose wind this time



This is the decesion tree for bootstrapped dataset 3.

TOPIC NAME : DAY : TIME : DATE : / /

lets predict :

Day 13 →

outlook	Temp	humidity	wind	Play
Rain	hot	high	weak	yes

Decesion tree 1 : No

decesion tree 2 → yes

decesion tree 3 → no

yes = 1

NO = 2

The final result of the random forest mode

is No.